

P7.11. The precharged circuit of Figure P7.11 is a 2-input AND gate. Draw the waveforms at nodes F , X , and Y as a function of time. You can assume short rise and fall times (i.e., you do not need to calculate them), but the final value in each phase must be correct. Only the capacitances shown here need to be considered. All devices are minimum size. Both F and Y start out unknown. Use $0.13\text{ }\mu\text{m}$ parameters.

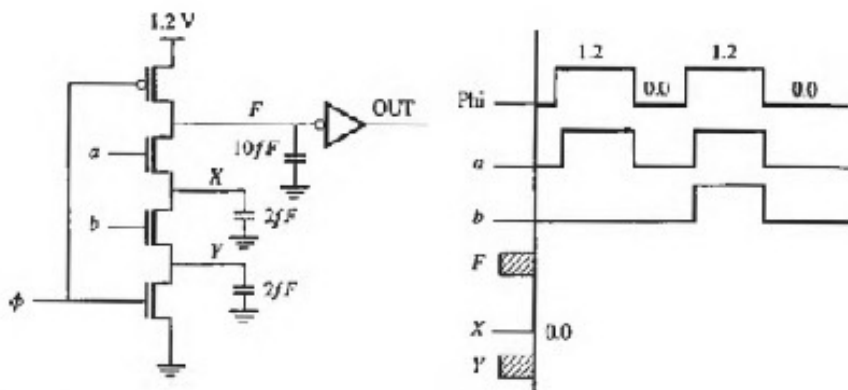
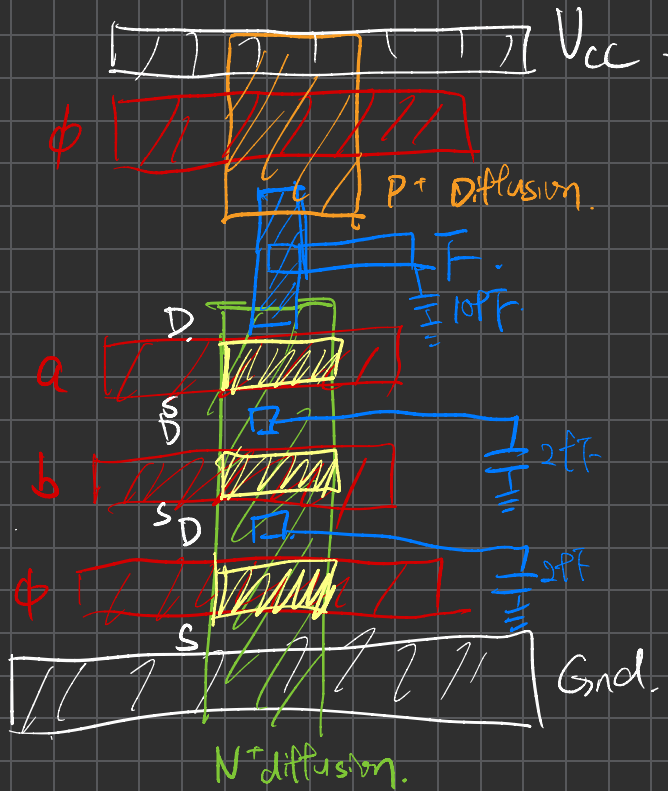


Figure P7.11

$$\text{For } a: V_{TN} = 0.4 + 0.2 (\sqrt{1.2 - V_{TN}} + 0.88 - \sqrt{0.88})$$

$$V_{TN} = 0.47\text{ V}$$

$$V_{T, \text{max}} = V_{DD} - V_{TN} = 1.2 - 0.47 = 0.73\text{ V}$$



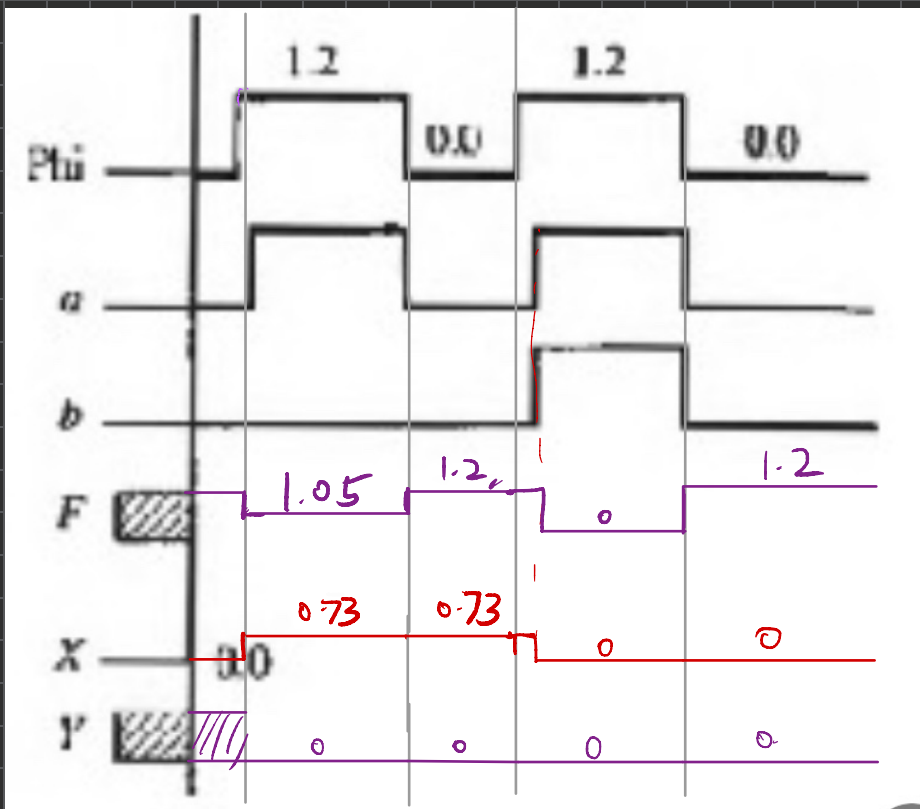
Before change: $Q_{Total} = 1.2 \times 10fF$

After change: $V_{xmax} \times 2fF + V_F \times 10fF = Q_{Total}$

$$0.73 \times 2f + 10f V_F = 12f$$

$$V_F = 1.05V$$

$$\Rightarrow V_x = 0.73V$$



Voltage at floated node is held.

P7.1. Compute all the voltages at all the nodes for the pass transistor circuit of Figure P7.1 including the body effect. Use $0.13 \mu\text{m}$ technology parameters. Assume that all nodes are initially at 0 V .

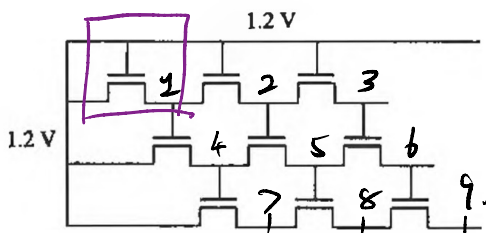


Figure P7.1

Note

$$1. \quad V_{TN} = V_{T0} + \gamma (\sqrt{(1.2 - V_{TN}) + 0.88} - \sqrt{0.88})$$

$$V_{TN} = 0.4 + 0.2 (\sqrt{(1.2 - V_{TN}) + 0.88} - \sqrt{0.88})$$

$$V_{TN} = 0.47 \text{ V}$$

$$V_1 = 1.2 - V_{TN} = 1.2 - 0.47 = 0.73 \text{ V}$$

$$V_2 = V_3 = V_1 = 0.73 \text{ V}$$

Note

$$4. \quad V_4 = V_1 - V_{TN,4}$$

$$V_{TN,4} = 0.4 + 0.2 (\sqrt{(0.73 - V_{TN,4}) + 0.88} - \sqrt{0.88})$$

$$= 0.43 \text{ V}$$

$$V_4 = V_5 = V_6 = 0.3 \text{ V}$$

$$V_4 = V_1 - 0.43 = 0.73 - 0.43 = 0.3 \text{ V}$$

$$V_4 < V_{TN} \Rightarrow V_7 = V_8 = V_9 = 0 \text{ V}$$